

# IFS-Light: An Interactive Framework for Single-view Face Relighting with both Facial and Lighting Consistency

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## A IMPLEMENTATION DETAILS

### A.1 Diffusion-based Relighting Module

Given an input image  $\mathbf{x}_0$ , a forward diffusion process is applied to obtain a noise map  $\mathbf{x}_{T=1000}$  under the condition  $h$ . The forward process of  $\mathbf{x}_{t+1}$  from  $\mathbf{x}_t$  is defined as:

$$\mathbf{x}_{t+1} = \sqrt{\alpha_{t+1}}g_{\theta}(\mathbf{x}_t, t, h) + \sqrt{1 - \alpha_{t+1}}\epsilon_{\theta}(\mathbf{x}_t, t, h), \quad (1)$$

where  $\epsilon_{\theta}$  is trained to predict the added noise  $\epsilon \sim \mathcal{N}(0, I)$  in  $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1 - \alpha_t}\epsilon$ , given the training image  $\mathbf{x}_0$ . We define  $\alpha_t = \prod_{s=1}^t (1 - \beta_s)$ , where  $\beta_t$  is the noise level at timestep  $t$  in the Gaussian diffusion process  $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t I)$ . The function  $g_{\theta}$  predicts  $\mathbf{x}_0$ , reparameterized from  $\epsilon_{\theta}$  as  $g_{\theta}(\mathbf{x}_t, t, h) = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \sqrt{1 - \alpha_t}\epsilon_{\theta}(\mathbf{x}_t, t, h))$ . The reverse process, conditioned on  $h'$ , is then applied to generate the relit image  $\mathbf{x}'_0$  by effectively reversing the forward diffusion process described in Eq. 1.

In the proposed consistent method, IFS-light, the conditional input is divided into two components to keep facial and lighting consistency: non-spatial conditions  $h = (c_o, b_o, a_t, i_t)$  and spatial conditions  $(SSG_o, LSG_t, M_o)$ . The non-spatial conditions act as standard conditional inputs in Eq. 1. Meanwhile, the non-spatial conditions are processed through a modulation network, which transforms them into spatial weights for the DDIM decoder. The DDIM decoder is based on the architecture proposed in [2], utilizing a modified UNet design that consists of a series of residual blocks interspersed with self-attention layers, as show in Fig. 1.

*Non-spatial Conditions.* We adopt the adaptive group normalization (AdaGN) for non-spatial conditions, as introduced in [9]. Given a feature map  $\mathbf{f}_{m_i} \in \mathbb{R}^{H_i \times W_i \times D_i}$ , AdaGN is defined as:

$$\text{AdaGN}_i(\mathbf{f}_{m_i}, c_o, b_o, a_t, i_t) = \mathbf{k}_i \left( t_i^s \text{GN}(\mathbf{f}_{m_i}) + t_i^b \right), \quad (2)$$

where  $\mathbf{k}_i = \text{MLP}_i^3(\text{Concat}(c_o, b_o, a_t, i_t)) \in \mathbb{R}^{D_i}$  is the output of a 3-layer MLP with SiLU activation [3]. Similarly,  $(t_i^s, t_i^b) \in \mathbb{R}^{2 \times D_i} = \text{MLP}_j^1(\gamma(t))$  are produced by a single-layer MLP with SiLU activation, a sinusoidal encoding function. Here, GN denotes standard group normalization.

*Spatial Conditions.* We utilize the Modulator network proposed in [9] to preserve spatial structure. The inputs to the Modulator network are  $SSG_o, LSG_t$ , and  $M_o$ . For each residual block  $j$  in the Modulator network, the output is denoted as  $\mathbf{m}_j \in \mathbb{R}^{H_j \times W_j \times D_j}$ , while the output of each residual block  $j$  in the DDIM is represented as  $\mathbf{o}_j$ . The adjusted output is then computed as  $\mathbf{o}'_j = \mathbf{o}_j \odot \tanh(\mathbf{m}_j)$ , where  $\odot$  denotes element-wise multiplication.

### A.2 Skin Tone Scaling Model

For our skin tone scaling model, we chose the RepViT [11] architecture because it is well-suited for training with limited data. We extracted frames from the videos provided in MST-E [10] and combined them with images to create the dataset. The data for the 10

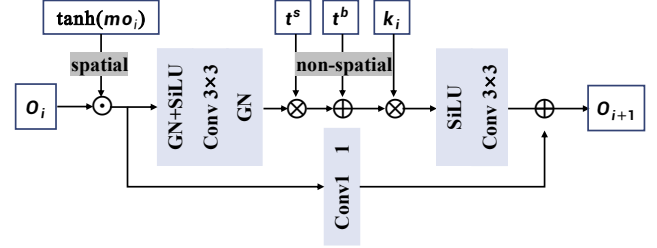


Figure 1: Diagram of non-spatial and spatial conditions injecting into one of the residual block in our conditional DDIM.

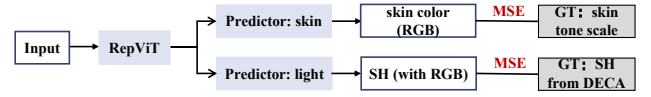


Figure 2: Pipeline of skin tone scaling model.

skin tone scales is divided proportionally into a training set (2508 images) and a testing set (667 images). To ensure accurate skin tone prediction, we employ a dual-branch training approach, as illustrated in Fig. 2. Initially, we extract features using RepViT and then split them into two branches: one for skin tone prediction and one for lighting color prediction. The skin tone branch uses skin tone grading as the ground truth (GT) and predicts skin tone values in RGB. The lighting color branch uses SH values predicted by DECA [4] as the GT and predicts SH values along with RGB values. We use Mean Squared Error (MSE) as the loss function and stop the training once the skin tone branch converges. During inference, only the skin tone is predicted. The angular error results of the skin tone scaling model across 10 different scales are shown in Table 1, demonstrating that our model can accurately predict skin tone under various lighting conditions. **The corresponding codes and the trained models are in the Code and Model folder of the supplementary material.**

### A.3 Training Losses

To ensure consistency in the relit images, our neural network is optimized using a set of carefully designed training losses:

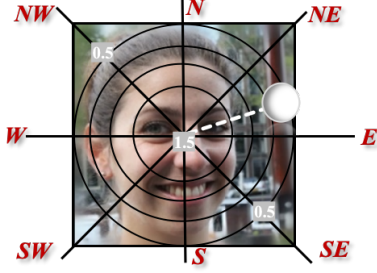
*Identity loss ( $L_{id}$ ):* To maintain identity consistency, we use the cosine similarity of identity features between the generated portrait and the input image.

*Reconstruction loss ( $L_{rec}$ ):* To ensure content fidelity, we employ the MSE loss between the generated portrait and the input image.

*Simplified loss ( $L_{simple}$ ) [9]:* To improve noise prediction and denoising, we introduce a simplified loss, whose parameters are

**Table 1: Results for the Angular Error of the Skin Tone Scaling Model across 10 Scales. The best is in bold.**

| Methods           | MST-1 ↓ | MST-2 ↓ | MST-3 ↓ | MST-4 ↓       | MST-5 ↓ | MST-6 ↓ | MST-7 ↓ | MST-8 ↓ | MST-9 ↓ | MST-10 ↓ | Avg ↓  |
|-------------------|---------|---------|---------|---------------|---------|---------|---------|---------|---------|----------|--------|
| Skin tone scaling | 0.19 °  | 0.13 °  | 0.32 °  | <b>0.05 °</b> | 0.62 °  | 1.17 °  | 1.85 °  | 2.63 °  | 0.47 °  | 0.93 °   | 0.93 ° |

**Figure 3: Visualization of the intensity and direction derived from the light ball.**

the same as Eq. 1:

$$L_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \|\epsilon_{\theta}(\mathbf{x}_t, t, h, M_{\theta}) - \epsilon\|_2^2, \quad (3)$$

The total loss is defined as:

$$L_{\text{total}} = L_{\text{rec}} + L_{\text{simple}} + \lambda L_{\text{id}}. \quad (4)$$

## A.4 Training Details

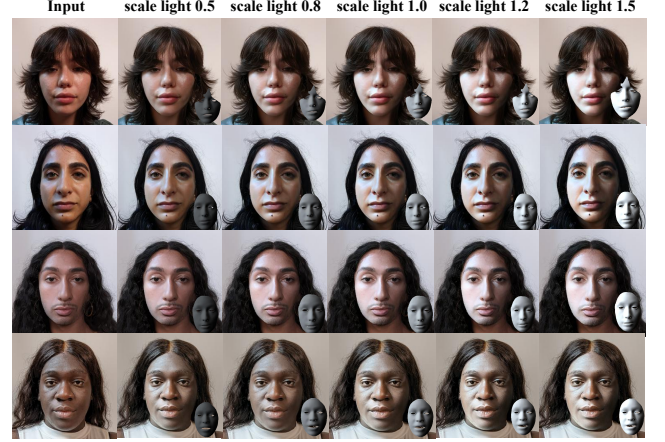
We normalized the training images to  $[-1, 1]$ , and precomputed their encodings from DECA [4], ArcFace [1], face parsing [12], Hou et al. (2022) [6], DiffAE [2] and the proposed skin tone scale model (details are in Sec.3.2.1 of the main text). We trained our DRM using training hyperparameters in Tab. 2.

**Table 2: Model Hyperparameters for DRM**

| Parameter            | DRM             |
|----------------------|-----------------|
| Base channels        | 128             |
| Channel multipliers  | [1, 1, 2, 3, 4] |
| Attention resolution | [16, 8]         |
| Batch size           | 16              |
| Diffusion step       | 1000            |
| Learning rate        | 1e-4            |
| Noise scheduler      | Linear          |
| Optimizer            | AdamW           |

## B DIY LIGHT

The interactive way of IFS-Light allows users to adjust the position of light ball for flexible control over light intensity and direction, making it easier to achieve the desired relighting results. As shown in Fig. 3, we visualize the intensity and direction derived from the light ball. The intensity decreases as the distance between the light ball and the center reduces, in conjunction with the predefined eight angles. We also demonstrate relighting results using our IFS-Light to adjust the lighting scale in Fig. 4. As the scaling factor increases,

**Figure 4: Relighting results with adjusted lighting scale factors. Note that input images are from the MST-E dataset, and the reference lighting uses the same Rembrandt lighting.**

the lighting intensity in the relit scene correspondingly intensifies. We further provide additional interactive relighting results in Sec. C. *Additionally, we provide a recorded video of using IFS-Light to perform relighting to demonstrate our interface and workflow in the Application folder of supplementary.*

## C ADDITIONAL RESULTS

In addition to the quantitative, qualitative, and ablation results presented in the main text, we also provide the following additional results here:

- Fig. 5 illustrates facial consistency across different views of the same subject, comparing results with Difareli [9].
- Fig. 6 distinguishes the varying roles of background conditions in comparison to Difareli [9].
- Fig. 7 demonstrates lighting consistency under various lighting conditions of the same subject.
- Fig. 8 shows qualitative results on Multi-PIE dataset [5].
- Fig. 9 outlines qualitative results on FFHQ dataset [8].
- Fig. 10 presents qualitative results on MST-E dataset [10].
- Fig. 11 indicates qualitative results of the ablation study on facial and lighting consistency conditions.
- Fig. 12 presents relighting results using our application with different lighting controls.

### C.1 Additional Facial Consistency Comparison

We compare our method with the diffusion-based method, Difareli [9], focusing on facial detail consistency. As illustrated in Fig. 5, we showcase relighting results across different views of the same subject using the same reference. Difareli tends to produce an overall yellowish skin tone, whereas our method achieves a more natural

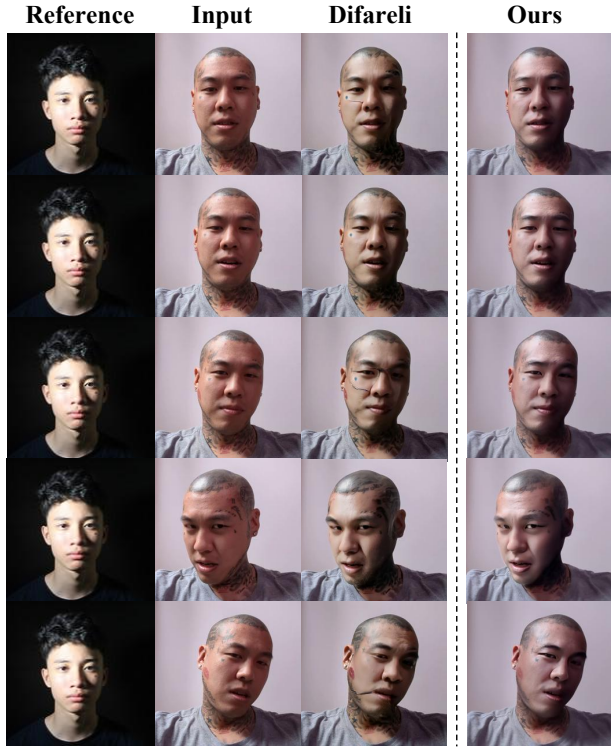


Figure 5: Results across different views of the same subject.



Figure 6: The role of background conditions.

and consistent skin tone across various views. Additionally, Difareli occasionally introduces artifacts such as black lines on the face and alters tattoos. In contrast, our method preserves more detailed and complete facial features, resulting in superior facial consistency.

It’s worth noting that the treatment of background exclusion in our method differs from that in Difareli, as shown in Fig. 6. In Difareli, background segmentation is used to extract background conditions as a conditional input. The results labeled “w/o bg” in Difareli indicate that this conditional input was omitted from the network. In contrast, our method employs a spatial encoder to generate a detail mask that excludes the background, which is still used as a conditional input. As demonstrated in the last image, our IFS-Light achieves consistent background appearance.

## C.2 Additional Lighting Consistency Comparison

We compare our method with the diffusion-based method, Difareli [9], with a focus on lighting detail consistency. As shown in Fig. 7,

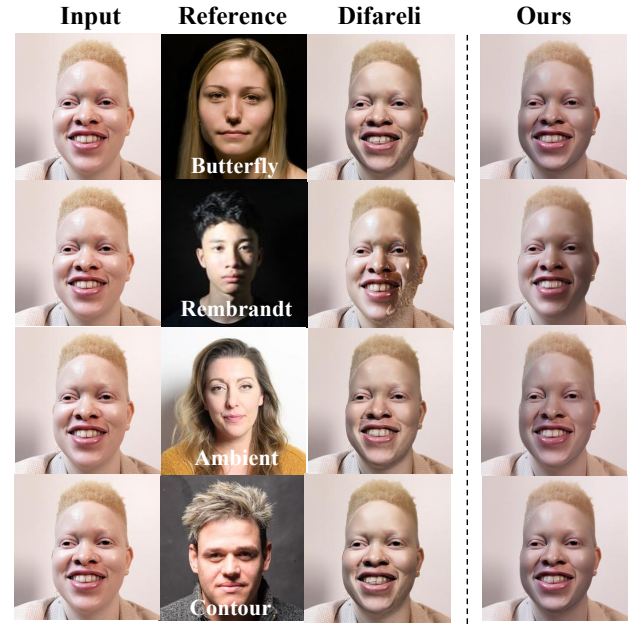


Figure 7: Results under various lighting of the same subject.

we present relighting results of the same subject under professional lighting conditions. Difareli struggles to handle scenarios involving both soft and hard shadows, such as Rembrandt lighting. In other cases, it fails to produce smooth transitions between shadowed and illuminated areas. In contrast, our IFS-Light method excels in these situations, generating consistent lighting.

## C.3 Additional Qualitative Results

Further qualitative results on Multi-PIE [5], FFHQ [8] and MST-E [10] datasets in Fig. 8, Fig. 9 and Fig. 9. In these figures, our method is benchmarked against prior state-of-the-art (SOTA) methods, namely DPR [13], Hou et al. (2021) [7], Hou et al. (2022) [6] and Difareli [9]. DPR and Hou et al. (2021) frequently exhibit overexposure or underexposure due to inaccurate lighting intensity estimation, resulting in inconsistent skin tones and lighting conditions that fail to align with input and reference images. Difareli, on the other hand, often introduces artifacts, such as altered details or incomplete faces, in the relighting outputs. In contrast, our method excels in achieving consistent facial and lighting effects while preserving fine details. It effectively balances specular highlights and hard shadows, maintaining facial identity, skin tones, and hair texture with remarkable fidelity, softness, and high-frequency detail. *For more details, please refer to the Visualization Comparison folder of the supplementary material.*

## C.4 Additional Ablation Results

In Fig. 11, we provide ablation visualizations on the Multi-PIE dataset, focusing on facial and lighting consistency.

*Facial Consistency Conditions.* Models w/o skin tone or w/o SSG fail to effectively control skin tones, often resulting in inconsistent



white or yellowish skin tones. Regarding details, models w/o base features or w/o detail mask struggle to preserve fine details, such as glasses (the white box). In contrast, the full model demonstrates its ability to maintain skin tones and preserve intricate details.

*Lighting Consistency Conditions.* Model w/o shadow introduces artifacts, such as altering hair color. Model w/o LSG produces results nearly identical to the input portraits, lacking enhanced lighting adjustments. The full model excels in removing hard shadows (highlighted by the white box) and achieving lighting consistency.

## C.5 Additional Interactive Results

Figure 12 presents additional relighting visualizations generated by our IFS-Light. The workflow consists of three key steps: 1) inputting source images, 2) interactively adjusting lighting conditions through mouse-controlled light ball manipulation, and 3) generating relighting results with corresponding lighting references. Our IFS-Light framework enables this flexible relighting way through its interactive lighting control system. **For more details, please refer to the Application folder of the supplementary material.**

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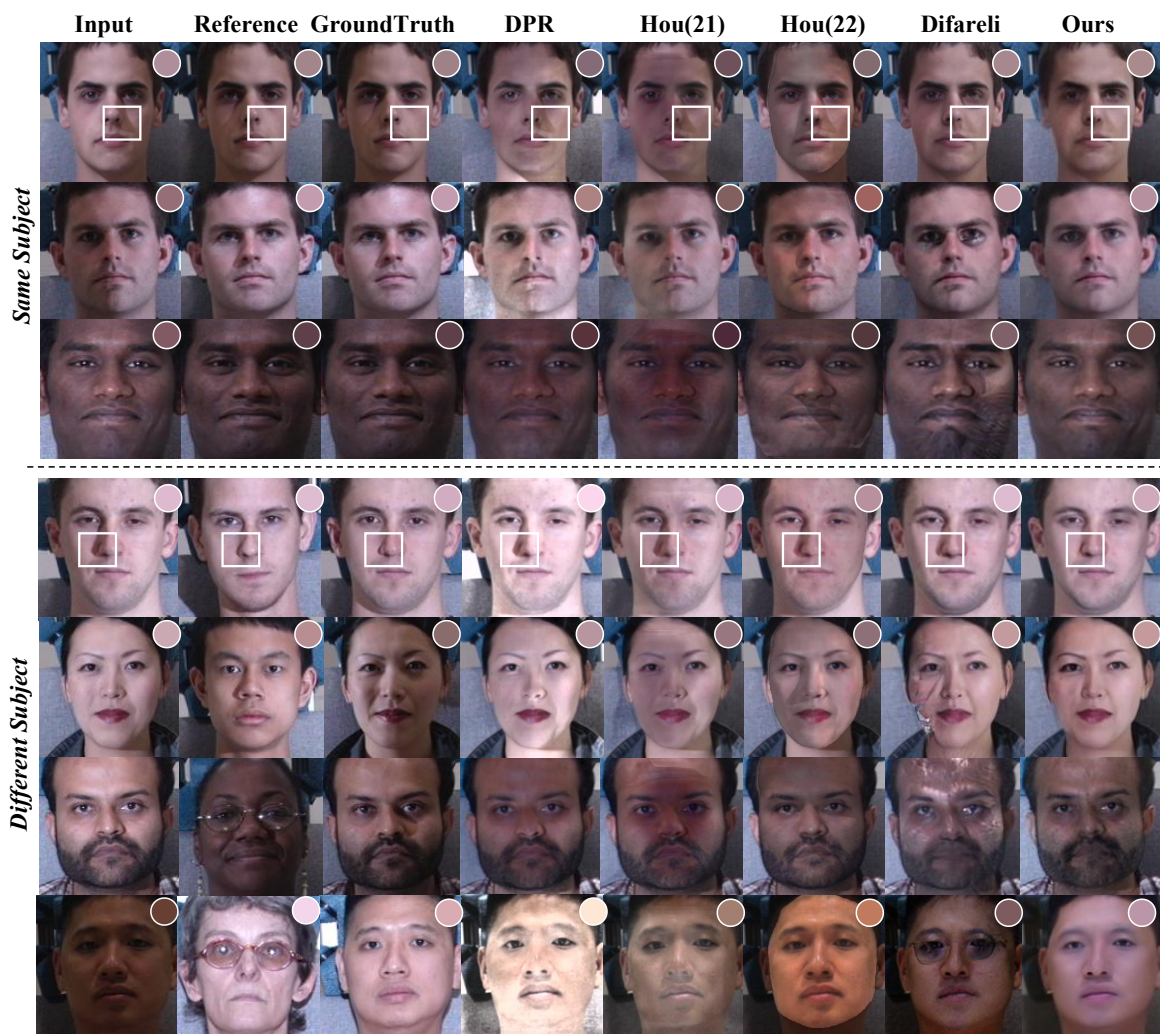


Figure 8: Relighting results on Multi-PIE dataset. Skin tone is indicated in the top-right corner, while the white boxes show specific details.



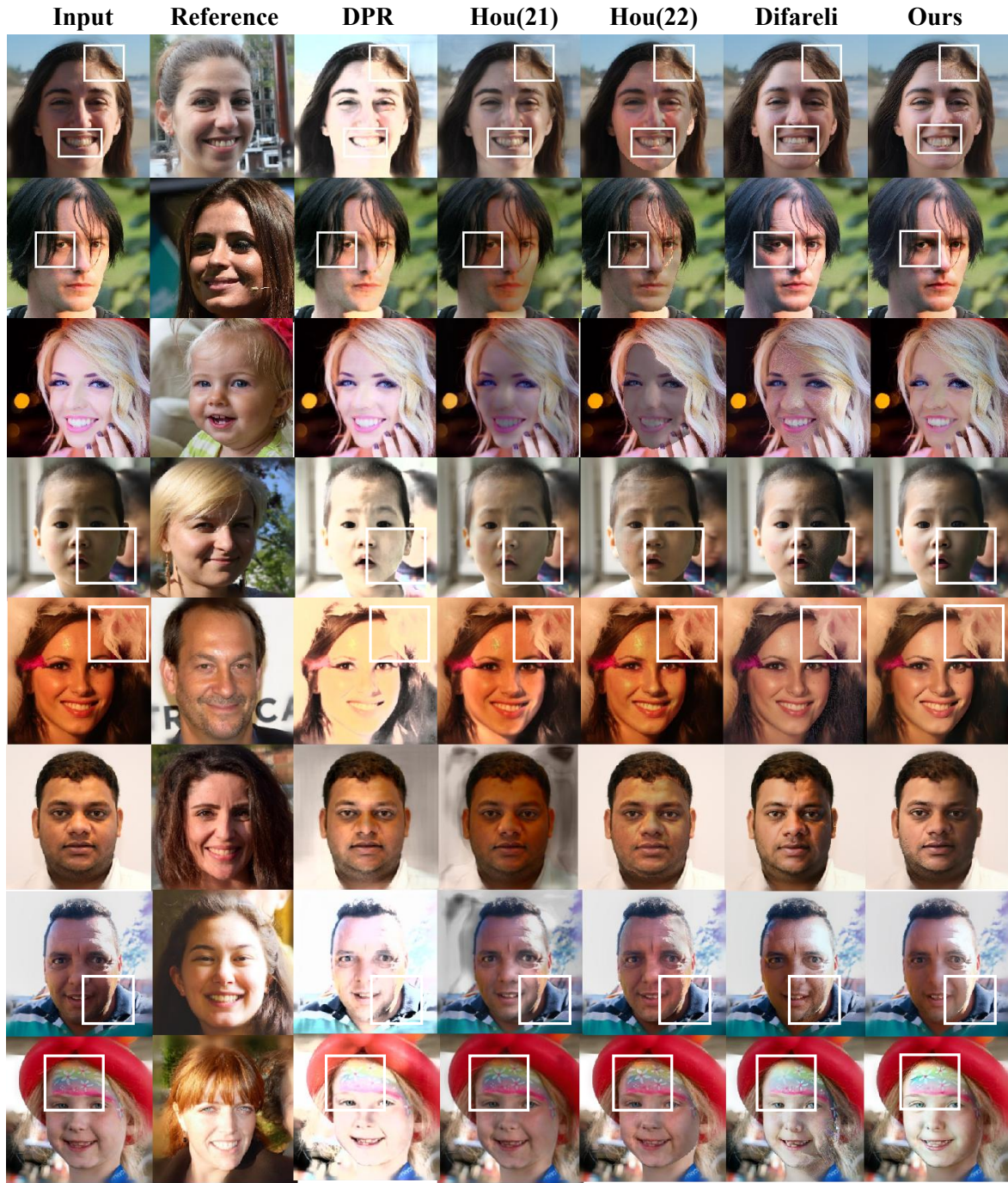


Figure 9: Relighting results on FFHQ dataset. Skin tone is indicated in the top-right corner, while the white box highlights specific details.



Figure 10: Relighting results on MST-E dataset. Skin tone is indicated in the top-right corner. Note that the reference images of MST-E experiments are from Flickr.



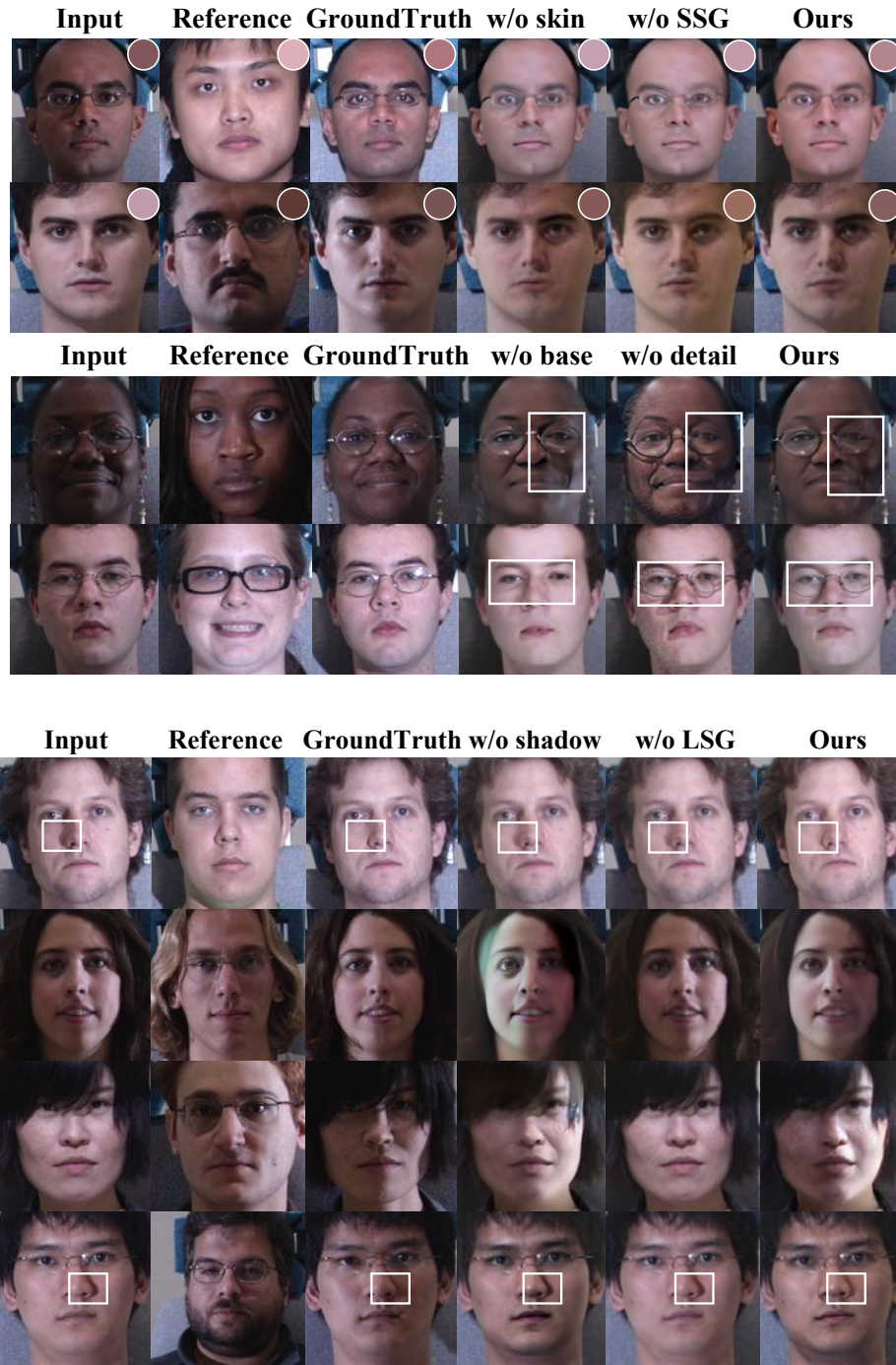


Figure 11: Ablation results on Multi-PIE dataset with facial and lighting consistency conditions. Skin tone is indicated in the top-right corner, while the white box highlights specific details.



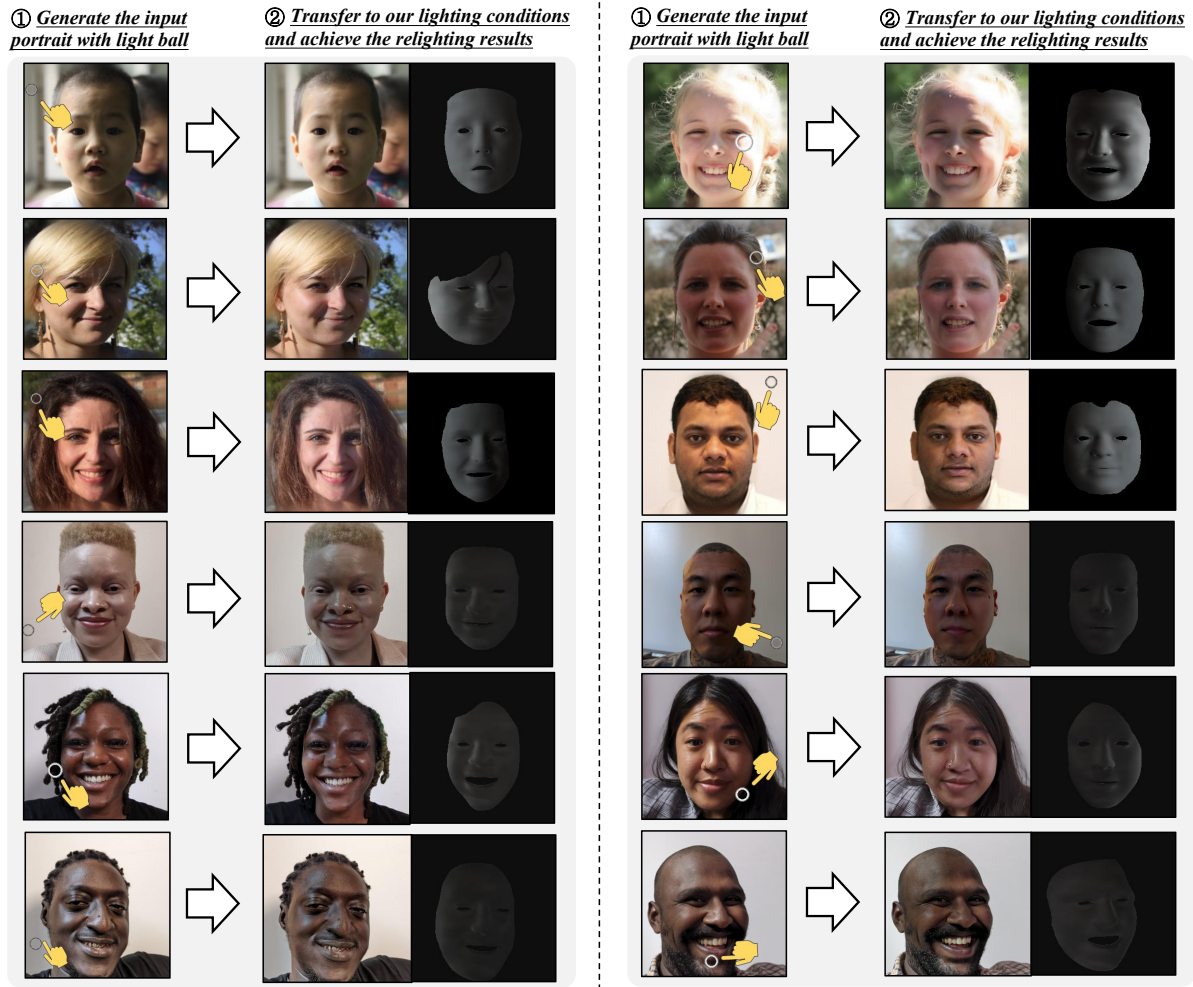


Figure 12: Relighting results using our application. Note that the input images are from FFHQ or MST-E.